Kidney Exchange with Multiple Donors*

Anand Siththaranjan†

September 17, 2025

Abstract

This paper studies the problem of finding efficient, strategy-proof, and weak-core stable mechanisms for kidney paired exchange under cycle size restrictions. We allow patients to have multiple donors, where their private information is their strict preference over the donor used. We show that no desirable pairwise mechanism exists, but under mild conditions constructively show that such a mechanism exists when cycles of length three are allowed. Our results leverage the structure of blood type compatibility to overcome classic impossibility results due to cycle size restrictions, and we provide intuition to this end. When considering the number of transplants made, imposing strategy-proofness while allowing multiple donors introduces a trade-off when compared to a complete information single-donor baseline. Nevertheless, we show through simulation using US population data that in most cases there is a relative increase in the number of transplants. With sufficiently many patients with more than one donor this can range from approximately 5 to 20%.

1 Introduction

One of the major successes of modern market design was the implementation of kidney paired exchange (KPE). When suffering from kidney failure, the most effective medical treatment is transplantation of a living kidney from a healthy donor. However, even when patients have a willing donor, they may not be able to donate due to biological compatibility

*We thank Haluk Ergin for his support throughout this project, as well as helpful comments from Tak-Huen Chau, Jianmeng Lyu, Alvin Roth, Chris Shannon, and Quitzé Valenzuela-Stookey.

†University of California, Berkeley, Department of Electrical Engineering and Computer Science, Berkeley, 2594 Hearst Ave, Berkeley, CA 94720, USA. E-mail: anandsranjan@berkeley.edu

restrictions. KPE emerged as a solution to this problem by identifying sets of patient-donor pairs where patients could exchange their willing donors so that all patients can receive a compatible donation from a donor different than their own (Roth et al., 2004).

From the perspective of a patient, this requires them to identify potential donors of their own first. In general, a patient may have multiple donors to choose from, and may in fact have preferences over them. These preferences could reflect objective medical features such as the risk of donation for different donors, or other subjective features such as personal relationships, familial obligations, and more. For example, some donors may have a family whereas others don't, and in the case where they are the sole provider for their family then it may be difficult to take time off from work due to post-operative care. Real-world implementations of KPE mechanisms do allow agents to bring multiple donors, however they tend to not solicit preference. A key concern with this is that it can lead to strategizing on the part of patients. For example, patients may choose to not bring certain less preferred donors to the mechanism before attempting to be matched using a preferred donor first.

From the perspective of a designer, this behaviour is not ideal. We would prefer patients to bring all their available donors to the exchange rather than gradually over time so that we can generate efficient outcomes. Achieving efficiency also requires patients to be able to express preferences over their donors. This motivates our work in trying to design an efficient, strategy-proof and individually rational mechanism when donors can be strictly ranked. Of key importance in the practice of KPE is ensuring that small cycles are used, as transplantation surgeries for KPE tend to be simultaneous. This paper aims to bridge a gap in the current literature on market design for KPE, which generally does not allow for general preferences over a patient's set of donors. First we consider the case where an exchange must be pairwise, finding an impossibility result in this environment. Pairwise exchanges are commonly one of the first studied settings for new organ exchange mechanisms with cycle-constraints (Roth et al., 2005b; Ergin et al., 2020, 2017), and this impossibility result contrasts with positive results when agents are indifferent over all their donors (Roth et al., 2005b). In and of itself, this may not be surprising considering the literature on house exchange with cycle-constraints (Balbuzanov, 2020; Kamada and Yasuda, 2025). However the results of the latter do not directly imply results in our environment. In particular, when allowing for two-and-three cycle exchanges, we find a positive result under mild assumptions. In the discussion section of the paper, we will develop intuition for how the structure of our problem is amenable to cycle-constraints.

A key metric by which KPE designers evaluate solutions on is the number of matches made. Imposing strategyproofness will generally lead to the first-best outcome not being implementable, whereas allowing multiple donors provide more opportunities to be matched over a single-donor baseline. Through simulations on US population data for various probabilities that a patient may have multiple donors, we compare of our mechanism with this baseline that has perfect information. We generally find weak improvements in terms of the relative increase in the number of matches made, and with moderate to high probabilities this ranges from approximately 5 to 20%. We conclude with a discussion of additional properties of our mechanism, and comments on the environment we study.

Outline. This paper is organized as follows. Section 2 describes the model of multi-donor kidney paired exchange, as well as the desired properties of a mechanism. Section 3 shows the non-existance of desirable pairwise exchange mechanisms. Section 4 studies the existence of desirable three-way exchanges under certain assumptions. Section 5 provides a simulation analysis of the potential gains from our mechanism over a single-donor complete information baseline. Section 6 provides a discussion of various features of the the studied environment and our solution.

1.1 Literature Review

The advent of kidney exchange (Rapaport, 1986) as a focus of the market design literature begins with Roth et al. (2004), which formulates the exchange problem as a variant of traditional house exchange problems. By viewing donors as houses and patients as agents, they built on Gale's Top Trading Cycles (TTC) algorithm to allow for chains initiated by altruistic (or deceased) donors. Subsequent papers incorporated practical constraints such as cycle size, and evaluated their theoretical properties (Roth et al., 2005b, 2007a) and simulation evidence (Roth et al., 2005a) as a foundation for the promise of this approach. We focus on considering strategic incentives when multiple donors are allowed. Though some algorithms do allow multiple donors to be listed (Abraham et al., 2007), and it has been noted that doing so can be useful for improving the performance of kidney exchange programs (Biró et al., 2019), an analysis of this setting and design solutions is missing from the current literature.

A common consideration in these mechanisms is to incorporate the preferences of patients over kidneys received (Roth et al., 2004; Nicolo and Rodriguez-Alvarez, 2017), which can may be derived from features such as the donors age and general health. In this paper we focus on preferences over donors used on behalf of the patient rather than on kidneys.

Though such preferences can be incorporated, our results on strategic incentives are only on the choice of patients to reveal, and truthfully rank, their donors. Though both problems can be viewed from the perspective of house exchange, whereas the classic approach tends to have strict ordinal preferences, our approach contains many indifferences. Because of this, we can bypass usual impossibility results due to cycle-constraints (Balbuzanov, 2020; Kamada and Yasuda, 2025), and achieve positive results given certain assumptions.

Our work contributes to a growing literature on incorporating new *modalities* that enrich the space of preferences that patients can express. By this we mean different ways in which patients and donors can participate in organ exchanges. Recent work includes incorporating ABO-desensitization in kidney exchange (Andersson and Kratz, 2020), and left- and right-lobe liver exchange (Ergin et al., 2020). Early work by this paper's author also explores abstractly how to incorporate new modes, serving as a framework for scaling current markets to novel technologies (Siththaranjan, 2025)[1]. This work can view allowing multiple donors, and expressing preferences over them, as having different donation modes.

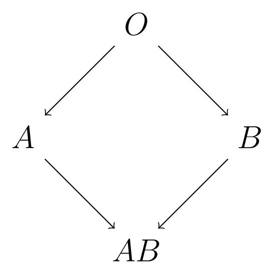
Introducing new modalities can encourage participation in organ exchanges by provide more opportunities. Other approaches to improving participation in kidney donation generally include providing incentives in the form of future priorities for donors (Sönmez et al., 2020; Kessler and Roth, 2012; Li et al., 2023; Kim and Li, 2022), incorporating compatible patient-donor pairs to participate in exchanges (Sönmez and Ünver, 2014), and allowing for donation-before-receipt[2] in exchange for improved priority (Akbarpour et al., 2025).

Our paper provides some insight into the literature on house exchange with indifferences. As TTC, even with tie-breaking, cannot generally handle such environments, approaches such Top Trading Absorbing Sets (TTAS) algorithm (Alcalde-Unzu and Molis, 2011) build on the intuition of TTC. Though such an approach cannot generally guarantee satisfying a cycle-constraint, as we require, the possibility result we achieve suggests that certain structures on preferences may allow these approaches to be compatible. Though non-conclusive, we provide some insight into this.

In practice, current algorithms do not tend to solicit preference information beyond quality cutoffs on kidneys to be received. In the context of our work, it is often the case that preferences over donors brought by a patient cannot be expressed. As algorithms used in practice tend solve a maximum-weighted cycle packing problem (Abraham et al., 2007;

[1] In this work, I consider a simple model of multi-donor kidney exchange as well. However, I assume preferences are derived from objective medical risk, and thus the preference between donors (but not the outside option) is known.

[2] This refers to allowing donors to donate a kidney before their patient receives, contrary to having simultaneous donation.



(a) Compatibility Graph

$X \leq_B Y$	O	A	B	AB
O	✓	✓	✓	✓
A		✓		✓
B			✓	✓
AB				✓

(b) Table description

Figure 1: Blood type compatibility.

Biró et al., 2009; Anderson et al., 2015), differences between donors that affect whether they are used in an exchange comes down to the weights assigned to them. A result of this is that truth-telling may not be optimal, and Pareto efficiency not achieved[3]. It has been observed that, at least on the part of hospitals, there are strategic incentives to misreport patient-donor pairs to centralized exchanges in order of matching them internally (Ashlagi and Roth, 2014)[4]. Match offers are also declined at rates of 25 to 35% in many prominent exchanges (Ashlagi and Roth, 2021a), and though this is partly due to the difficulty in pre-specifying acceptable exchanges, there is potential for strategic incentives to be playing a role. Our approach explicitly deals with these issues by identifying an efficient, stable and strategyproof mechanism, however there is a tradeoff as it may lose optimality according to general maximum-weight objectives. Nevertheless, we see our result as providing insight into how new algorithms could be structured to incorporate incentives.

2 Model

Let $\mathcal{B} = \{A, B, AB, O\}$ be the set of blood types. A donor with blood type b can donate to a patient with blood type b' if $b \leq_{\mathcal{B}} b'$, where $\leq_{\mathcal{B}}$ is a reflexive, transitive partial order as depicted in Figure 1. The biological characterization is as follows. A and B are antigens, and your blood type specifies if you have either antigen. For example, AB means you have both, whereas O means you have neither. A donor can donate to a recipient if whenever a recipient is missing an antigen, then the donor is also missing the antigen. Some basic

[3] We provide more discussion on this in Section 6.

[4] The reason for this is that hospitals generally aim to maximize the number of their own patients who receive a transplant, and thus may match those that are easier to do so internally and leverage the centralized exchange to match those that are harder to match. This can lead to inefficiencies and loss in aggregate welfare as compared to the first best. See Ashlagi and Roth (2014) for a more in-depth analysis and discussion.

observations of this order tells us important details about the ease or difficulty by which a blood type can perform as a donor or recipient with regards to facilitating a transplant:

1. O is a universal donor: $O \leq_{\mathcal{B}} b$ for all $b \in \mathcal{B}$
2. AB is a universal recipient: $b \leq_{\mathcal{B}} AB$ for all $b \in \mathcal{B}$
3. A and B are incompatible: $A \not\leq_{\mathcal{B}} B$ and $B \not\leq_{\mathcal{B}} A$
4. AB cannot donate to other types: $AB \not\leq_{B} b$ for $b \in \mathcal{B} \setminus \{AB\}$
5. O cannot receive from other types: $b \not\leq_{B} O$ for $b \in \mathcal{B} \setminus \{O\}$

An agent is a patient-donor tuple $i \in \mathcal{I}$, where we denote $\tau_i = P - D_1, D_2, D_3, D_4, D_5 \in \mathcal{T} = \cup_{n=1}^{5} \mathcal{B} \times (\mathcal{B} \cup \{\emptyset\})^n$ to be the type of an agent. Let $\mathcal{D}_i = \{D_1, D_2, D_3, D_4, D_5\} \cap \mathcal{B}$. The type τ_i contains the following observable and private information:

1. P is the observable blood type of a patient
2. $\mathcal{D}_i \cup \{\emptyset\}$ are blood types of donors or the outside option $\emptyset$, which is referring to using no donor and thus not participating in any exchange. We observe the elements of $\mathcal{D}_i$ but not its order.
3. The ordered set composed of $\mathcal{D}_i \cup \{\emptyset\}$ is the private preference ordering $\succ_i$ on donors and the outside option $\emptyset$. We assume that $D_1 \succ_i D_2 \succ_i D_3 \succ_i D_4 \succ_i D_5$ for $D_k \in \mathcal{D}_i \cup \{\emptyset\}$, though we may not be able to identify D_k

We can assume that patients list up to 4 donors with no repeated blood types[5], any donor dis-preferred to the outside option will nevertheless be listed, and patients may be blood type compatible with their donors. We will find these assumptions to be without loss in our mechanism. We will also use the notation $\mathcal{I}_{P-D_1*}$ to refer the set of agents with blood type P for their patient, D_1 for their most-preferred donor, and $*$ means that they may have further donors. Furthermore, $n_{P-D_1*} = |\mathcal{I}_{P-D_1*}|$. Similarly we define $\mathcal{I}_{P-D_1,D_2*}$ where D_2 is the agent's second favourite donor, and so forth. Using the same notation, we succinctly write an agent's type $\tau_i = P - D_1*$ to emphasize that the D_1 donor, their most-preferred donor, is most relevant.

A **paired exchange environment** is $\mathcal{E} = (\mathcal{I}, \mathcal{T})$. Given such an environment, we use the notation $i \to_b j$ for $i, j \in \mathcal{I}$ and $b \in \mathcal{B}$ to mean that i **donates to** j **using a** b **donor**. We only consider feasible donations, that is a donation $i \to_b j$ such that

[5] If a patient has a donor with repeated blood type, we will find that in our mechanism, the patient need only bring their most preferred donor among those of the same blood type.

1. i has a donor with blood type b, i.e. $b \in \mathcal{D}_i$

2. b can donate to the patient in j, i.e. $b \leq_{\mathcal{B}} P_j$

An environment $\mathcal{E}$ induces an edge-labeled directed multi-graph we refer to as a **compatibility graph**: $G_{\mathcal{E}} = (\mathcal{I}, F)$, where the set of labeled edges $F \subseteq \mathcal{I} \times \mathcal{I} \times \mathcal{B}$ is such that $(i, j, b) \in F$ if and only if $i \rightarrow_b j$. An exchange $E = \{(i_1^l \rightarrow_{b_1^l} \cdots \rightarrow_{b_{k_l}^l} i_{k_l}^l)\}_{l \in L}$ is a set of vertex-disjoint cycles indexed by L, where a single cycle $l \in L$ is $i_1^l \rightarrow_{b_1^l} \cdots \rightarrow_{b_{k_l}^l} i_{k_l}^l$ made up of distinct agents. If the blood type used in a donation is not defined, we assume it is inferred from the preferences of the agent from which it is outgoing. In particular, for $i_k^l \rightarrow i_{k+1}^l$ we assume they use their most preferred donor b that can facilitate that donation (i.e. $i_k^l \rightarrow_b i_{k+1}^l$). An n-exchange is an exchange E such that for all cycles $l \in L$, we have that $k_l \leq n$. We refer to the set of exchanges by $\mathbb{E}$, and the set of n-exchanges by $\mathbb{E}^n$. In the special case where $n = 2$, we refer to an exchange as a **matching** and define $\mathbb{M} = \mathbb{E}^2$. We use the notation $E(i)$ to refer to the blood type of the donor used by i in E, and $\emptyset$ otherwise. Formally, $E(i) = b$ if there exists $j \in \mathcal{I}$ such that $(i, j, b) \in E$, and $\emptyset$ otherwise. Note that this is well-defined as exchanges are vertex disjoint, hence there is a unique donor used and agent that is donated to.

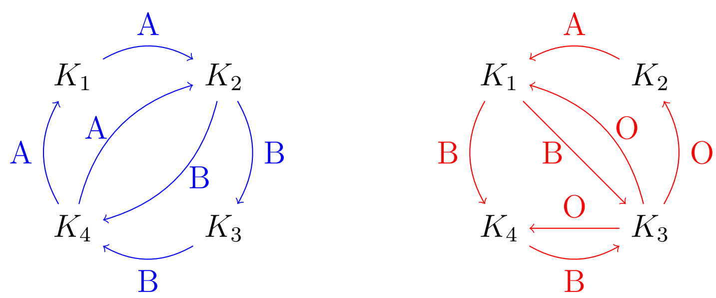
We now details some useful operations and definitions applied to exchanges. Let `MaxMatch`(A) compute a maximum exchange of size at most 3 among all agents $A \subseteq \mathcal{I}$. When we say that an agent i of type $X - YZ*$ **drop** their Y donor, we transform their type to be $X - Z*$ and eliminate their donor Y from being considered. Let Π be a priority order over $\mathcal{I}$, and $\Pi|_A$ be the same order restricted to $A \subseteq \mathcal{I}$. For a set of exchanges M and M', let the operation $M \leftarrow M'$ mean to add M' to M, and to remove all agents in M' from $\mathcal{I}$. Finally, we say that $S \subseteq \mathcal{I}$ is **matchable** if there exists $E \in \mathbb{E}^3$ such that for all $i \in S$, $E(i) = \mathcal{D}_i^1$. When a donor of an agent i is dropped prior to being added to a set S, matchability is with respect to their available donors at the time at which i was added. We define the function `Match`(M, S) to output an exchange E such that $M \subseteq E$ and all agents $i \in S$ are matched via their top donor in E.

The following example provides a simple two-donor setting while illustrating and comparing different types of exchanges:

Example 1 (Two Donor Example)**.** Consider the environment $\mathcal{E}$ given by Figure 2a, with compatibility graph for D_1 and D_2 edges given by Figures 2b and 2c respectively. The set of agents can be written by $\mathcal{I} = \{K_i\}_{i=1}^4$ where $K_1 = A - A, B$, $K_2 = A - B, A$, $K_3 = B - B, O$, and $K_4 = B - A, B$.

$\mathcal{I}$	P	D_1	D_2
K_1	A	A	B
K_2	A	B	A
K_3	B	B	O
K_4	B	A	B

(a) Environment $\mathcal{E}$



(b) $G_{\mathcal{E}}$ with D_1

(c) $G_{\mathcal{E}}$ with D_2

Figure 2: An example environment with two donors.

An example of a pairwise exchange would be $E_1 = \{(K_2 \to K_4)\}$, however a (larger) exchange like $E_2 = \{(K_1 \to K_2 \to K_3 \to K_4)\}$ has more agents participating with their preferred donor. Although K_2 and K_4 are exchanging with different agents in E_2 as compared to E_1, they are still using the same donor. Furthermore, there are agents that were not exchanging in E_1 (K_1 and K_3) who are now participating in E_2. E_2 is ideal in that all agents participate in an exchange, and they are able to use their favorite donor.

Consider $E_3 = \{(K_1 \to K_4 \to K_3 \to K_2)\}$. Similar to E_2, there are four agents participating in this exchange. However, these agents are using their least preferred donor. Thus we would intuitively prefer E_2 over E_3.

If instead K_1 had type $\tau_{K_1} = A - B, A$, that is they prefer their B donor over their A donor and thus have the opposite preference, we can compare $E_4 = \{(K_1 \to K_3 \to K_4)\}$ to E_3. The latter has strictly more agents in the exchange, however the K_1 would strictly prefer E_4 over E_3. $\triangle$

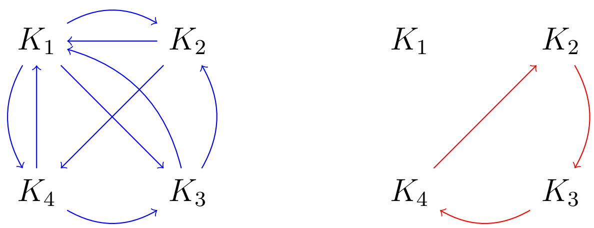
2.1 Mechanisms

Given an environment $\mathcal{E}$ where $n = |\mathcal{I}|$, an **exchange mechanism** is $\phi : \mathcal{T}^n \to \mathbb{E}$. ϕ is a **pairwise mechanism** if $\phi[\mathcal{T}^n] \subseteq \mathbb{M}$, and a **two-and-three-cycle mechanism** if $\phi[\mathcal{T}^n] \subseteq \mathbb{E}^3$. Let $\bar{\mathbb{E}}$ be the range of ϕ.

Consider some $\tau \in \mathcal{T}^n$ and $E = \phi[\tau]$. We say that E is

1. **individually rational** (IR) if for all $i \in \mathcal{I}$, $E(i) \succeq_i \emptyset$.

2. **Pareto efficient** (PE) if there does not exists $E' \in \bar{\mathbb{E}}$ such that for all $j \in \mathcal{I}$, $E'(j) \succeq_j E(j)$, and there exists some $i \in \mathcal{I}$ such that $E'(i) \succ_i E(i)$.



(a) G with D_1 (b) G with D_2

Figure 3: A candidate compatibility graph G, corresponding to Figure 1 of Gilon et al. (2019).

3. **weak-core stable** (WCS) if there does not exist $A \subseteq \mathcal{I}$ and exchange $E' \in \mathbb{E}$ such that for all $i \in A$, $E'(i) \succ_i E(i)$

A mechanism ϕ is PE, IR, and WCS if for every $\tau \in \mathcal{T}^n$, $E = \phi[\tau]$ is PE, IR, and WCS. A **deviation** for agent i from $\tau_i = P_i - D_1, \ldots, D_5$ is $\tau_i' = P_i - D_{k_1}, \ldots, D_{k_5}$, where $\{k_i\}_{i=1}^5$ is a permutation of $\{1, \ldots, 5\}$. A mechanism is **strategy-proof** (SP) if for all $\tau \in \mathcal{T}^n$, $i \in \mathcal{I}$ and deviations τ_i' from τ_i, we have that $\phi[\tau](i) \succeq_i \phi[\tau_i', \tau_{-i}](i)$. We say a mechanism is **desirable** if it is PE, IR, WCS, and SP.

3 Pairwise Mechanisms

We begin with studying the most minimal setting of short cycle mechanisms - that is pairwise exchange mechanisms. This is a common setting of study in many market design approaches to kidney exchange (Roth et al., 2005b; Sönmez and Ünver, 2014; Ergin et al., 2020). We ask whether there exists a desirable pairwise exchange mechanism for all environments. The following result shows that this does not hold.

Proposition 1. *There exists an environment where there is no desirable pairwise exchange mechanism.*

The structure of this proof identifies a counter-example with three agents whose patients and donors have A and O blood types only[6]. In fact, it only requires leveraging three of the four criteria for desirability - in particular, strategy-proofness, Pareto efficiency, and individual rationality - and shows that this restricts the different feasible matches at different type profiles. We then find a contradiction should these criteria simultaneously hold.

[6] Given that O and A are the two most common blood types, this is not an edge case example.

Importance of Compatibility Structure. An important feature of this problem which has been ignored in past work is on incorporating the structure of biological compatibility. In particular, Gilon et al. (2019) have previously studied kidney exchange with multiple donors and the existence of pairwise mechanisms. By example, they aim to show the impossibility of a desirable mechanism under a pairwise cycle constraint. However, as we show below, their example is not possible under the standard blood type compatibility model. In particular, the structure of blood type compatibility restricts the set of feasible trades. Let $G = (\mathcal{I}, H)$ be a candidate compatibility digraph where $H \subseteq \mathcal{I} \times \mathcal{I}$. Note that this compatibility graph does not specify the blood type of the donor used in a directed edge. We say that G is **rationalizable** if there exists an environment $\mathcal{E}$ such that for $G_{\mathcal{E}} = (\mathcal{I}, F)$, we have that $H = \mathbf{proj}(F) = \{(i,j) \in \mathcal{I} \times \mathcal{I} | \exists b \in \mathcal{B},\ (i,j,b) \in F\}$. In other words, there exists a choice of blood types for patients and donors that make these matches as exactly the feasible set of matches.

Proposition 2. *The compatibility graph G given in Figure 3 is not rationalizable.*

This result is illustrative about the importance of accounting for the blood-type compatibility structure. Though they find an impossibility result conjectured to be in the same setting as ours, it does not provide an underlying blood type structure to rationalize their example. It may be the case that a richer model of compatibility in kidney exchange, beyond those standard in the literature (Roth et al., 2005b), may allow for such a matching to be feasible. Nevertheless, we find that our restriction to blood-type compatibility is informative about the degree to which an impossibility result holds. We will see in the subsequent section that accounting for blood type compatibility is essential for a positive result. Although richer models may change the structure of feasible matches, our result highlights that at the most essential and basic level of compatibility, all hope for desirable mechanism is not lost. Thus our proposed mechanism can be seen as describing an intuitive structure that has desirable qualities in an ideal setting, and may generalize to finer, richer models.

In light of this, we can view the result of Gilon et al. (2019) as a comment on the existence of desirable mechanisms when any trade is feasible and preferences over the outside option can be arbitrary. This may be useful in other areas of interest, which are the beyond the scope of this work.

4 A Triadic Mechanism

We now explore the existence of desirable triadic mechanisms, i.e. one that uses cycles of size at most three. We consider various assumptions either for simplicity in the design of the mechanism, or based in realistic assumptions on the distribution of agents.

Our first assumption allows us to simplify the design of our mechanism:

Assumption 1. $n_{AB-B*} + \min\{n_{AB-A*}, n_{A-O*} + n_{A-B*}\} \leq n_{B-AB*} \leq n_{AB-B*} + n_{AB-O*} + \min\{n_{AB-A*}, n_{A-O*} + n_{A-B*}\}$.

This assumption is based in observations of the empirical distribution of kidney patients blood types and their PRA levels, which allows us to infer the probability of entering a random pair of agents being tissue-type incompatible and thus entering the exchange[7]. We leverage this assumption to allow certain agents to be "exhausted" at certain stages of the mechanism. In doing so, this weakens the strategy-proofness constraint from taking advantage of our efficiency requirement in later stages of the algorithm. For example, all $AB-A$ agents are "exhausted" by $B-AB$, meaning that they are in an exchange together. As such, a Pareto improvement for an agent i that strictly prefers matching with $AB-A$ would require rematching $B-AB$ in a different exchange. We structure our algorithm to make this impossible without violating efficiency, and thus limits i's options. In doing so, we make it such that there are less ways for i to deviate, thus mitigating their latitude for strategizing.

As AB donors can only donate to AB patients, this assumption implies that it is without loss of generality that all agents only have non-AB donors, and have at most three donors. We provide further discussion of this assumption in Section 6. This next assumption is common in the literature, as stated in Roth et al. (2007a), which is motivated by the intuition that there are many $O-A$ and $O-B$ pairs due to blood-type incompatibility between patient and donor[8] and the high proportion of O blood types[9].

Assumption 2 (Long-side of the Market)**.** *At least one of each type in $O-b$ for $b \in \{A, B\}$ is unmatched in any feasible matching. If $X \rightarrow Y$, then $n_{X-Y*} \geq n_{Y-X*}$*

This final assumption is for simplicity of the mechanism, similar to that in Roth et al. (2007a) except we preclude the possibility of no $b - b*$ agents:

[7]See Appendix B.1 for more details.

[8]This leverages the idea that patients tend to enter the exchange after trying to use their donors but failing due to compatibility.

[9]Approximately 48% in the US.

Assumption 3. *There are at least two $O-O*$, $B-B*$, and $A-A*$.*

This, as well as the previous, assumption can be view as a *large market* assumption, as they are likely to hold in sufficiently large markets.

Without loss of generality, we assume $n_{A-B*} \geq n_{B-A*}$, and our algorithm is analogous in the case where $n_{A-B*} < n_{B-A*}$[10]. Consider the following mechanism: $M = \emptyset$.

1. Match $\mathcal{I}_{B-B*}$ amongst each other: $M \leftarrow \texttt{MaxMatch}(\mathcal{I}_{B-B*})$.

2. All agents $i \in \mathcal{I}_{A-*}$ drop AB.

3. Match $\mathcal{I}_{A-B*}$ and $\mathcal{I}_{B-A*}$: for $i \in \Pi_{\mathcal{I}_{A-B*}}$.

   (a) If $n_{B-A*} \neq 0$, then choose some $j \in \mathcal{I}_{B-A*}$ and do $M \leftarrow \{(i \to j)\}$.

   (b) Else, exit this for loop.

4. Match $\mathcal{I}_{B-AB*}$:

   (a) For $i \in \Pi_{B-AB*}$: If $n_{AB-B*} > 0$, find some $j \in \mathcal{I}_{AB-B*}$ and do $M \leftarrow \{(i \to j)\}$.

5. Define $S = \mathcal{I}(M) \cup \mathcal{I}_{AB-*} \cup \mathcal{I}_{B-AB*} \cup \texttt{Top}(n_{AB-A*}, \mathcal{I}_{A-B*}|\Pi)$. (Commit $AB-A$ and $A-B$ to match with $B-AB$, and $AB-B$ as well)

6. All agents $i \in \mathcal{I}_{A-*} \setminus S$ drops B.

7. Serial Dictator-like procedure with agents $O-A*$, $O-B*$ and $O-AB*$: for $i \in \Pi_{\mathcal{I}_{O-A*} \cup \mathcal{I}_{O-B*} \cup \mathcal{I}_{O-AB*}}$

   (a) i chooses favorite of $j \in \mathcal{I}_{A-O*} \cup \mathcal{I}_{B-O*} \cup \mathcal{I}_{AB-O*} \cup \{\emptyset\}$ such that $S \cup \{i, j\}$ is matchable. In such a case $S \leftarrow S \cup \{i, j\}$. If i prefers matching with $O-O$ than these options, then they exit.

   (b) If not possible, then continue.

8. For $i \in \mathcal{I}_{\mathcal{I}_{O-*}}$, i drops their A, B and AB donors (if any).

9. Match $\mathcal{I}_{O-O*}$ amongst each other: $M \leftarrow \texttt{MaxMatch}(\mathcal{I}_{O-O*})$.

10. Match $\mathcal{I}_{A-A*}$ amongst each other: $M \leftarrow \texttt{MaxMatch}(\mathcal{I}_{A-A*})$.

[10] When showing strategyproofness of a mechanism, we will account for the potential that a misreport may change this order.

11. Match $\mathcal{I}_{AB-AB*}$ amongst each other: $M \leftarrow \texttt{MaxMatch}(\mathcal{I}_{AB-AB*})$.

12. Return $\texttt{Match}(M, S)$

For intuition, we now informally describe the algorithm and provide commentary. In the first step, we match all $B-B*$ agents amongst themselves and thus allow them to have their best option. We then require all A patients to drop their AB donor, hence they are unable to match using them. This step foreshadows that these agents will in fact be unable to use their AB donor in a Pareto improving exchange, and thus it is without loss to drop them from consideration. As there are more $A-B*$ than $B-A*$, we use a priority order to let $A-B*$ be matched with $B-A*$. Agents matched at this step get their best option of those that remain. Observe that thus far, $B-*$ agent that have matched have done so with their top donor. Excluding AB donors, the same can be said of $A-*$ agents. Step four matches $B-AB*$ with a $AB-B*$ agent, which exhausts the latter by assumption and ensures that all $B-*$ get their ideal match. At this point, we can conclude that no $B-*$ can be strictly improved.

At step 5, we identify a group of agents S that contains the previously matched agents as well as $AB-*$, $B-AB*$ and some $A-B*$ agents. We can guarantee these agents to be matched via their top available donor, and note that the only potential strategic behavior may come from $A-B*$. However, because we use a priority order to select agents and they agents are guaranteed to match via any other non-AB donor they may have[11], then there is no incentive to deviate. We treat this set S as a *promise* that these agents will be matched using the donor specified at the time they were added. At this point, we have exhausted the oppportunities for $A-*$ agents to match using their B donor. Thus in step 6, such agents drop their B donor should they have one.

The remaining agents that are non-trivial to match are $O-*$ agents. However these agents, apart from $O-O*$, do not participate in an exchange with each other. As such, we can interpret these agents as the "agents" of serial dictator, and feasible exchanges involving certain over-demanded agents (like $X-O*$ for $X \in \{A, B, AB\}$) as "objects" they can choose. By leveraging matchability as a constraint, we can ensure that exchanges are feasible while preserving earlier promises. We do not directly finalize exchanges as agents may be indifferent between multiple exchanges. This approach can be viewed as similar to serial dictator with indifferences, where agents at their turn of the algorithm filter the set of possible exchanges by removing those that are not their most preferred.

[11] This follows because it is "easy-to-match" $A-O$ and $A-A$ agents.

The final steps of the algorithm match agents in groups with patient-donor pairs of the same blood type as this is the only option remaining. As we have matched various agents as well as created promises for agents to use certain agents, the last step returns a feasible matching that respects these constraints. Because we have leverage matchability throughout the algorithm we specific matches were not specified, this step is guaranteed to return an exchange.

Throughout the algorithm, we have gradually exhausted an agent's best option. By reasoning similarly to properties of sequential procedures like serial dictatorship, we can see that this helps to mitigate strategic behavior. As exhaustion of options for an agent i to use a preferred donor implies that previous agents that were already matched must be re-matched differently, we can see how efficiency may be established. The following theorem formalizes our observations into properties of our mechanism:

Theorem 1. *This mechanism is desirable.*

Our proof follows similarly from our observations, however requires leveraging the stated assumptions to establish when certain exchange opportunities for an agent's donor have been exhausted.

5 Simulations

We compare our mechanism with a random priority order to a maximum matching baseline where agents only bring their top donor[12]. We see this baseline as a setting where agents take the simple strategy of only bringing their top donor, rather than a more complicated strategy based on the equilibrium distribution of outcomes. As most papers study a single donor environment, or do not account for general donor preferences, we aim to highlight the practicality of mechanisms that do account for these factors.

We randomly sample patients and donors from a common blood type distribution (see Figure 1), except we assume there are no AB patients for simplicity of simulation[13]. We consider different probabilities that an agent may have one, two, or three donors. We only allow patients to have a top donor with the same blood type as them if they are tissue-type incompatible. We use the same probabilities as in Roth et al. (2007a), where

[12] Such a mechanism can be found in Roth et al. (2007a).

[13] This allows us to simplify the check of matchability, and means that AB donors are not needed. As AB patients form a small fraction of the population, we don't anticipate this having a substantial effect on our experiment.

Parameter	**Value**
Blood Type Probabilities (ABO)	O: 0.41 A: 0.36 B: 0.23
Percentage Reactive Antibodies (PRA)	L: 0.70 M: 0.20 H: 0.10
Crossmatch Probability	L: 0.05 M: 0.45 H: 0.90

Table 1: Simulation configuration parameters (rounded two decimal places from Roth et al. (2007b)). We do not include AB members of the population, and normalize ABO probabilities accordingly. The PRA parameters specifies the proportion of the population with a given level, and the crossmatch probability for that level is an independent likelihood of an agent being tissue-type incompatible with any given donor.

for each patient we sample a PRA level from $\{L, M, H\}$ and then randomly sampling with the relevant probabilities whether they are tissue-type compatible. Furthermore, we only consider distributions of agents that satisfy our assumptions in Section 4. Figure 4 shows the average relative increase in our mechanism compared to the baseline when there are 200 agents for different donor number probabilities. We average our results over 1000 random trials.

Improvements over the baseline are dependent on the number of agents with more than one donor. We can see in Figure 4 that there are approximately little, if any, losses from imposing strategyproof-ness, and relative increases from approximately 5 to 20% when at least 60% of agents have more than one available donor. This suggests that careful construction of multi-donor mechanisms can have promising improvements over exchanges that only allow a single donor to be listed. Though in practice multiple donors may be allowed in certain systems, the lack of clear incentives to report all of them can result in agents leveraging a simple strategy of reporting their most preferred donor only.

Assumption on AB patients. We now provide further justification for the assumption to not include AB patients in our simulation analysis. Firstly, in the United States, AB blood types make up approximately 4% of the population (Roth et al., 2007b). In general, it is a very rare blood type. Secondly, as AB patients can receive from any blood type,

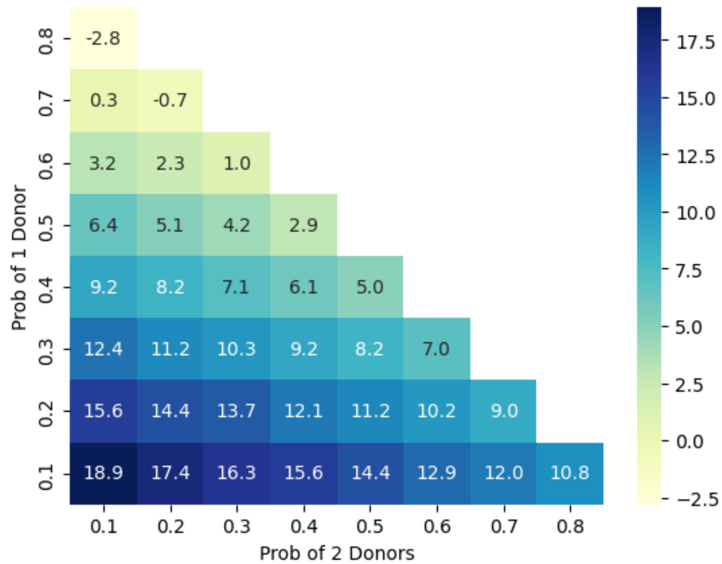


Figure 4: (%) Relative Increase for $n = 300$ given different probabilities of the number of donors. Averaged over 1000 random trials. The probability of 3 donors is given by $1 - \mathbb{P}(1 \text{ donor}) - \mathbb{P}(2 \text{ donors})$.

we can note that the probability that their top donor is also one they are tissue-type incompatible with is approximately 20%[14]. Hence they are unlikely to enter a paired exchange mechanism, and our assumption can be seen as implying that we do not consider a group of patients that make up less than 1% of the population. As such, the gains from incorporating agents with AB patients is likely to be minimal, and these patients are also those that tend to be easy to match.

6 Discussion

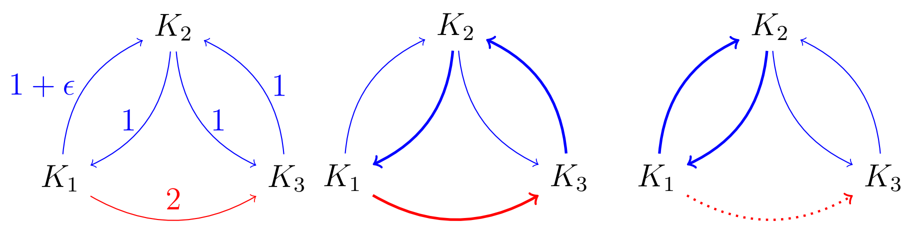
Incentives and (In)Efficiency in the Status Quo. Current algorithms tend to not account for preferences. Rather they usually take a set of reported donors, construct a compatibility graph with weights on directed edges and cycles, and solve a maximum-weight cycle packing problem with constraints on the size of cycles[15]. Weights can be

[14]This can be calculated using the probabilities in Roth et al. (2007a). See Appendix B.1 for more details.

[15]Formally, the problem of maximum-weight cycle packing is as follows. Consider some directed graph $G = (V, E)$. Let n be the restriction on the size of a cycle, and $w_{i,j}$ be an edge weight. Then for exchange $E = \{(i_l^k)_{k=1}^{k_l}\}_{l \in L}$ define $w(E) = \sum_{l \in L} \sum_{k=1}^{k_l} w_{k,(k+1) \bmod k_l}$. Then a solution to this problem solves $\max_{E \in \mathbb{E}^n} w(E)$.

$\mathcal{I}$	P	D_1	D_2
K_1	A	B	A
K_2	B	A	$\emptyset$
K_3	A	B	$\emptyset$

(a) Environment $\mathcal{E}$



(b) $G_{\mathcal{E}}$ with weights (c) Exchange E_1 (d) Exchange E_2

Figure 5: An example environment with weights on the compatibility graph where agents have an incentive to strategically misreport in the maximum-weight exchange. Bold lines refer to donations used in the specified exchange, and dotted lines refer to unreported donors.

chosen based on features that are donor-specific, such as match quality[16]. The following simple example highlights how patients can have strategic incentives to misreport their set of feasible donors[17].

Example 2 (Strategic Deviation)**.** Consider the environment $\mathcal{E}$ given by Figure 5a, with weights on edges being shown in Figure 5b. Assume $\epsilon > 0$ small. If all agents reported their full set of feasible donors, the maximum-weight cycle packing is given by exchange $E_1 = \{(K_1 \to K_2 \to K_3)\}$, as shown in Figure 5c. However, if K_1 did not report their less preferred donor, then the maximum-weight cycle packing is $E_2 = \{(K_1 \to K_2)\}$ (shown in Figure 5d). Hence K_1 has incentive to deviate from truth-telling. $\triangle$

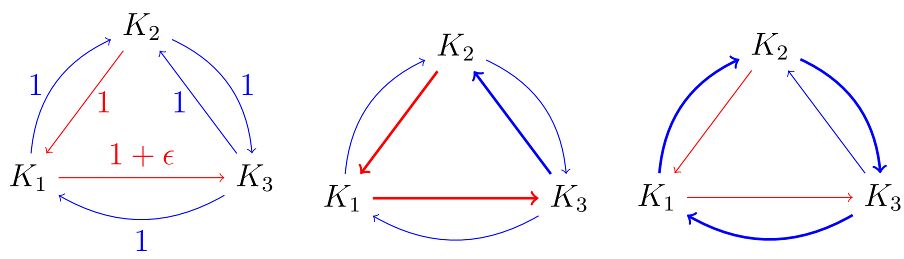
In practice, individual patients tend to not have sufficient information to be able to perfectly strategize. Nevertheless, people may still try to take "safe strategies" like only reporting their top donor. If they are unmatched, then in a later iteration of the algorithm they may bring both. As designers, we would prefer agents to bring all their donors in order to achieve our objective as best as we can, while minimizing the cognitive burden associated with strategizing.

If there were beliefs on the likelihood of an agent having a second donor, then alternative choices of exchanges may be beneficial. For example, in the previous example, consider $\epsilon > 0$ sufficiently small. If we choose $K_2 \leftrightarrow K_3$ if K_1 did not report their second donor, and $K_1 \to K_2 \to K_3$ otherwise, then there may be higher expected value while being strategy-proof. Though such a model is outside the scope of this paper, it provide some motivation

[16] Ashlagi and Roth (2021b) describe an example of this by the "life years gained from [a] transplant", which may depend on features such as the donor's age and health.

[17] Note that examples can be found when weights are all identical, which corresponds to the case of transplant maximization. See the Appendix for more details. [TODO]

$\mathcal{I}$	P	D_1	D_2
K_1	A	A	B
K_2	A	B	A
K_3	B	A	$\emptyset$



(a) Environment $\mathcal{E}$ (b) $G_{\mathcal{E}}$ with weights (c) Exchange E_1 (d) Exchange E_2

Figure 6: An example environment with weights on the compatibility graph where the maximum-weight exchange is inefficient. Bold lines refer to donations used in the specified exchange.

for considering strategy-proofness as a criteria in designing paired kidney exchanges.

Furthermore, the following example shows how weight-maximization conflicts with Pareto efficiency.

Example 3 (Inefficiency)**.** Consider the environment $\mathcal{E}$ given by Figure 5a, with weights on edges being shown in Figure 5b. Assume $\epsilon > 0$ small. The maximum-weight cycle packing is given by exchange $E_1 = \{(K_1 \to K_3 \to K_2)\}$, as shown in Figure 6c. This is an inefficient exchange, whereas the Pareto dominating efficient exchange is $E_2 = \{(K_1 \to K_2 \to K_3)\}$ in Figure 6d. △

Though Pareto efficiency is often a desirable criteria in the general market design literature, in kidney exchange the focus is primarily on metrics such as average transplant quality and number of transplants (Roth et al., 2007a). Whether this should come as a cost of efficiency is a question beyond the authority of the author and the scope of this paper, but is one that should be addressed socially. We see our algorithm as providing new possibilities that satisfy different criteria to current algorithms, so that all options can be appropriately evaluated and compared.

How Many Donors Does One Need? Patients are faced with the need to solicit donors in order to participate in these exchanges. The task of doing so can be time-consuming and emotionally challenging for patients and potential donors alike. As such, we would want to characterize properties of our mechanism that allows us to say how few donors a patient can bring without altering their outcome in the mechanism. Formally, we say that a mechanism is k**-donor equivalent** for $A \subseteq \mathcal{I}$ if for any pair $i \in A$, their outcome in the mechanism is the same than if they truncated their preference to their top

k alternatives.[18] In other words, the mechanism is independent of the donors specified outside of their top k. As there are few AB patients, and thus having an AB donor is generally known to be difficult to use in an exchange, we will assume that there are no AB patients to have a more informative result:

Observation 1. *Assume that there are no AB patients, i.e. $\nexists i \in \mathcal{I}$ such that $P_i = AB$. Then the mechanism is*

1. 1*-donor equivalent for* $\mathcal{I}_{B-*}$,
2. 2*-donor equivalent for* $\mathcal{I}_{A-*}$, *and*
3. 3*-donor equivalent for* $\mathcal{I}_{O-*}$.

This result is useful in that certain patients need not spend more effort than necessary to find donors outside of their top k, where k is given by the above proposition.

TTC and cycle-constraints. When there are no cycle-constraints, variants of TTC that account for indifferences can easily be applied to our problem by the following. Let preferences over donation modes induce preferences over agents where agent i is strictly preferred to agent j by agent k if k can donate to i using a donation mode that is strictly better than any donation mode that k can use to donate to j. Thus, approaches that solve house exchange problems when there are indifferences, such as the Top Trading Absorbing Sets (TTAS) algorithm in Alcalde-Unzu and Molis (2011), can be used.

However, in our environment we require exchanges to be composed of two or three way cycles, that is cycles in an exchange are of length at most three. It is known in the literature that imposing cycle-constraints makes it such that there is generally no mechanism that is efficient, strategy-proof, and individually rational for house exchange (Kamada and Yasuda, 2025). So why is it possible in this environment to find such a mechanism when there is a cycle constraint? The following result gives some intuition by showing that if there is a top trading cycle, then there is one that has length two (i.e. a 2-cycle):

Proposition 3. *Consider a graph where all agents point to their favourite feasible agent. If there is a cycle, then there must be a 2-cycle.*

[18] Formally, for a given preference $\succ_i$ over $\mathcal{M} \cup \{\emptyset\}$, denote the **$k$-truncated preference** $\succ_i^k$ such that there exists $m^l \in \mathcal{M}$ distinct where $m^1 \succ_i \cdots \succ_i m^{|\mathcal{M}|}$ and $m^1 \succ_i^k \cdots \succ_i^k m^k \succ_i^k \emptyset \succ_i^k m^{k+1}$. Then a mechanism ϕ is **k-donor equivalent** for A if for all preference profiles $\succ$ and for all agents $i \in A$, then $\phi(\succ) = \phi(\succ_i^k, \succ_{-i})$.

This result, and its proof, show that the structure of the compatibility relation and donor preferences induces a certain structure on preferences over agents that is somewhat compatible with the concept of a top trading cycle. Though this provides some intuition for why it is possible, note however that we cannot use the TTC algorithm itself due to indifferences, and some approaches that adapt TTC to allow for indifferences cannot clearly be adapted to leverage this 2-cycle existence property. For example, when cycles are executed in TTAS (Alcalde-Unzu and Molis, 2011), objects are only provisionally assigned and thus may be reassigned in later parts of the algorithm. A priority order is leveraged to break ties, and in general this may result in large cycles. Transplant maximizing exchanges require four cycles in general, and this can be the only efficient outcome in some settings[19]. Thus TTAS will select it. If we weakened our criteria to four cycles, applying TTAS and rematching agents according to the procedure of Roth et al. (2007a) will result in a feasible exchange. Furthermore, we can observe that this procedure preserves the incentive, efficiency, and stability properties of TTAS. However in practice, kidney exchanges are limited to cycles of size at most three (Ashlagi and Roth, 2021b)[20].

Preferences over Kidneys. Absent from our model are preferences over kidneys, which is commonly allowed for in the literature. As we primarily focus on incentives associated with donor preferences, we opt to exclude this aspect for simplicity and view this as an extension of models with 0-1 preferences for kidney exchange (Roth et al., 2005b). We contend that this model still provides useful insights on the timing of matching agents and how this relates to preferences and blood types. Furthermore, from the perspective of surgeons, it has been noted in past work that they recommend indifference over living organs due to the similarity in outcomes (Roth et al., 2005b; Yılmaz, 2011; Sönmez and Ünver, 2014). For example, we can consider the age of a donor as a metric for health. Figure 4 in Terasaki et al. (1995) shows similar graft survival rates over a three year period for donors under the age of 50. Nevertheless, when preferences are observable, as sometimes assumed in the literature due to the preferences for healthier kidneys, we may be able to incorporate them using a similar approach as in Ergin et al. (2020).

[19]Roth et al. (2007a) give an example where there is a four cycle induced by $AB - O \to O - A \to A - B \to B - AB$. Note that adding $b - b*$ agents for $b \in \mathcal{B}$ as needed allows this example to satisfy our assumption.

[20]Note that increasing a cycle for k to $k{+}1$ involves undertaking two additional operations simultaneously with the others.

7 Conclusion

In this work we consider the existence of desirable - that is Pareto efficient, weak-core stable, individually rational, and strategy-proof - mechanisms for paired kidney exchange with multiple donors when cycles have size constraints and agents have preferences over which donor of theirs is used. We find negative results in the case of pairwise mechanisms, and positive results when allowing cycles of size at most three under reasonable assumptions. Our result is surprising in light of known results on cycle-constraints (Balbuzanov, 2020; Kamada and Yasuda, 2025) and recent work on the same topic (Gilon et al., 2019), and our progress emphasizes the importance of leveraging biological compatibility and practical distributional assumptions in these problems. By further testing this mechanism in simulation using US population data, we can see reasonable improvements over single-donor baselines that motivate the practical viability of multi-donor mechanisms that account for preferences to ensure efficiency, maintain stability, and mitigate strategic behavior.

References

Abraham, David J., Avrim Blum, and Tuomas Sandholm (2007), "Clearing algorithms for barter exchange markets: enabling nationwide kidney exchanges." In *Proceedings of the 8th ACM conference on Electronic commerce*, 295–304, ACM, San Diego California USA, URL `https://dl.acm.org/doi/10.1145/1250910.1250954`. 3, 4

Akbarpour, Mohammad, Julien Combe, YingHua He, Victor Hiller, and Robert Shimer (2025), "Unpaired Kidney Exchange: Overcoming Double Coincidence of Wants without Money." 4

Alcalde-Unzu, Jorge and Elena Molis (2011), "Exchange of indivisible goods and indifferences: The top trading absorbing sets mechanisms." *Games and Economic Behavior*, 73(1), 1–16. 4, 19, 20

Anderson, Ross, Itai Ashlagi, David Gamarnik, and Alvin E Roth (2015), "Finding long chains in kidney exchange using the traveling salesman problem." *Proceedings of the National Academy of Sciences*, 112(3), 663–668. 5

Andersson, Tommy and Jörgen Kratz (2020), "Pairwise kidney exchange over the blood group barrier." *The Review of Economic Studies*, 87(3), 1091–1133. 4

Ashlagi, Itai and Alvin E Roth (2014), "Free riding and participation in large scale, multi-hospital kidney exchange." *Theoretical Economics*, 9(3), 817–863. 5

Ashlagi, Itai and Alvin E. Roth (2021a), "Kidney Exchange: An Operations Perspective." *Management Science*, 67(9), 5455–5478, URL `https://pubsonline.informs.org/doi/10.1287/mnsc.2020.3954`. 5

Ashlagi, Itai and Alvin E Roth (2021b), "Kidney exchange: An operations perspective." *Management Science*, 67(9), 5455–5478. 17, 20

Balbuzanov, Ivan (2020), "Short trading cycles: Paired kidney exchange with strict ordinal preferences." *Mathematical Social Sciences*, 104, 78–87. 2, 4, 21

Biró, Péter, Bernadette Haase-Kromwijk, Tommy Andersson, Eyjólfur Ingi Ásgeirsson, Tatiana Baltesová, Ioannis Boletis, Catarina Bolotinha, Gregor Bond, Georg Böhmig, Lisa Burnapp, et al. (2019), "Building kidney exchange programmes in europe—an overview of exchange practice and activities." *Transplantation*, 103(7), 1514–1522. 3

Biró, Péter, David F. Manlove, and Romeo Rizzi (2009), "MAXIMUM WEIGHT CYCLE PACKING IN DIRECTED GRAPHS, WITH APPLICATION TO KIDNEY EXCHANGE PROGRAMS." *Discrete Mathematics, Algorithms and Applications*, 01(04), 499–517, URL `https://www.worldscientific.com/doi/abs/10.1142/S1793830909000373`. 5

Ergin, Haluk, Tayfun Sönmez, and M Utku Ünver (2017), "Dual-donor organ exchange." *Econometrica*, 85(5), 1645–1671. 2

Ergin, Haluk, Tayfun Sönmez, and M Utku Ünver (2020), "Efficient and incentive-compatible liver exchange." *Econometrica*, 88(3), 965–1005. 2, 4, 9, 20

Gilon, Oren, Tal Gilon, and Assaf Romm (2019), "Kidney exchange with multiple donors." *Available at SSRN 3451470*. 9, 10, 21, 27

Kamada, Yuichiro and Yosuke Yasuda (2025), "Robust exchange under single-peaked preferences." *Working paper*. 2, 4, 19, 21

Kessler, Judd B and Alvin E Roth (2012), "Organ Allocation Policy and the Decision to Donate." *American Economic Review*, 102(5), 2018–2047, URL `https://pubs.aeaweb.org/doi/10.1257/aer.102.5.2018`. 4

Kim, Jaehong and Mengling Li (2022), "Optimal organ allocation policy under blood-type barriers with the donor-priority rule." *Theoretical Economics*, 17(1), 331–369, URL `https://econtheory.org/ojs/index.php/te/article/view/3969`. 4

Li, Mengling, Yohanes E Riyanto, and Menghan Xu (2023), "Prioritized organ allocation rules under compatibility constraints." *Games and Economic Behavior*, 141, 403–427. 4

Nicolo, Antonio and Carmelo Rodriguez-Alvarez (2017), "Age-based preferences in paired kidney exchange." *Games and Economic Behavior*, 102, 508–524. 3

Rapaport, Felix T (1986), "The case for a living emotionally related international kidney donor exchange registry." In *Transplantation proceedings*, volume 18, 5–9. 3

Roth, Alvin E, Tayfun Sönmez, and M Utku Ünver (2004), "Kidney exchange." *The Quarterly journal of economics*, 119(2), 457–488. 2, 3

Roth, Alvin E, Tayfun Sönmez, and M Utku Ünver (2005a), "A kidney exchange clearinghouse in new england." *American Economic Review*, 95(2), 376–380. 3

Roth, Alvin E, Tayfun Sönmez, and M Utku Ünver (2007a), "Efficient kidney exchange: Coincidence of wants in markets with compatibility-based preferences." *American Economic Review*, 97(3), 828–851. 3, 11, 14, 16, 18, 20, 33

Roth, Alvin E., Tayfun Sönmez, and M. Utku Ünver (2005b), "Pairwise kidney exchange." *Journal of Economic Theory*, 125(2), 151–188, URL `https://www.sciencedirect.com/science/article/pii/S0022053105001055`. 2, 3, 9, 10, 20

Roth, Alvin E, Tayfun Sönmez, and M. Utku Ünver (2007b), "Efficient Kidney Exchange: Coincidence of Wants in Markets with Compatibility-Based Preferences." *American Economic Review*, 97(3), 828–851, URL `https://pubs.aeaweb.org/doi/10.1257/aer.97.3.828`. 15

Siththaranjan, Anand (2025), "Multi-modal paired exchange." *Working paper*. 4

Sönmez, Tayfun and M. Utku Ünver (2014), "Altruistically unbalanced kidney exchange." *Journal of Economic Theory*, 152, 105–129, URL `https://linkinghub.elsevier.com/retrieve/pii/S002205311400057X`. 4, 9, 20

Sönmez, Tayfun, M. Utku Ünver, and M. Bumin Yenmez (2020), "Incentivized Kidney Exchange." *American Economic Review*, 110(7), 2198–2224, URL `https://pubs.aeaweb.org/doi/10.1257/aer.20170950`. 4

Terasaki, Paul I, J Michael Cecka, David W Gjertson, and Steven Takemoto (1995), "High survival rates of kidney transplants from spousal and living unrelated donors." *New England Journal of Medicine*, 333(6), 333–336. 20

Yılmaz, Özgür (2011), "Kidney exchange: An egalitarian mechanism." *Journal of Economic Theory*, 146(2), 592–618. 20

A Proofs

A.1 Proof of Proposition 1

Proof. Consider the following agents, that is patients with donors (that may or may not be feasible):

- $\{i, j, k\}$ each with an O patient and $\{O, A\}$ donors.
- l with an A patient and an O donor.

Observe that for any of $\{i, j, k\}$ to match with each other, it must be through their O donor, and to match with l can be through either donor. However to use their A donor, they must match with l. We will define a match by the agents participating in it, with their favourite donor used implicitly. For example (i, j) means that the donate to each other via their O donors. We say a match is valid if it is efficient and IR given the preferences considered.

Assume for contradiction that there is a desirable mechanism. Fix l's preference as $O \succ \emptyset$. Now consider the following preference profile $\succ_1$ for $\{i, j, k\}$:

$$i : A \succ \emptyset \succ O$$
$$j : A \succ \emptyset \succ O$$
$$k : A \succ \emptyset \succ O$$

Without loss of generality, let the valid allocation in the desirable mechanism match (i, l) together. Consider the following profile $\succ_2$:

$$i : O \succ A \succ \emptyset$$
$$j : A \succ \emptyset \succ O$$
$$k : A \succ \emptyset \succ O$$

Since (i, j) and (i, k) are not IR, then to maintain strategy-proofness we must have (i, l) match. Now consider the following $\succ_3$:

$$i : O \succ A \succ \emptyset$$
$$j : A \succ O \succ \emptyset$$
$$k : A \succ \emptyset \succ O$$

The valid matches are $\{(i,j),(k,l)\}$ and (j,l). However choosing the latter would violate strategy-proofness as there would be a profitable deviation for j to misreport in preference profile $\succ_3$ into $\succ_2$ where they go from being unmatched to being part of an individually rational match. Hence let the match here be $\{(i,j),(k,l)\}$. Now consider the following $\succ_4$:

$$i : O \succ \emptyset \succ A$$
$$j : A \succ O \succ \emptyset$$
$$k : A \succ \emptyset \succ O$$

Note that (j,l) is valid but not strategy-proof, as otherwise i will misreport from $\succ_4$ to $\succ_3$ and get from being unmatched to matched with j. Thus the only valid match is $\{(i,j),(k,l)\}$. Now consider the following $\succ_5$:

$$i : O \succ \emptyset \succ A$$
$$j : A \succ \emptyset \succ O$$
$$k : A \succ \emptyset \succ O$$

Note that (j,l) is valid but again not strategy-proof, as otherwise j in $\succ_4$ would misreport in $\succ_5$ to $\succ_4$ and go from (i,j) to (j,l), which is a strictly better outcome for them. Hence the only valid outcome is (k,l). Now consider the following $\succ_6$:

$$i : O \succ \emptyset \succ A$$
$$j : A \succ \emptyset \succ O$$
$$k : A \succ O \succ \emptyset$$

Observe if (k,l) is not the valid match chosen here, then k in $\succ_6$ will misreport to $\succ_5$ to get this match and thus strictly improve. Hence (k,l) is the outcome here. Now consider the following $\succ_7$:

$$i : O \succ A \succ \emptyset$$
$$j : A \succ \emptyset \succ O$$
$$k : A \succ O \succ \emptyset$$

There are two valid outcomes here: $\{(i,k),(j,l)\}$ and (k,l). Note that the former would not be strategy-proof, as otherwise we would have i misreport from $\succ_6$ to $\succ_7$ and thus go from being unmatched to being matched in a valid outcome. However the latter would also not be strategy-proof, as k in $\succ_2$ would misreport to $\succ_7$ and go from being unmatched to being matched in a valid outcome. Thus there is no way of choosing a valid match. Hence there is no desirable mechanism. $\square$

A.2 Proof of Proposition 2

Note that we adopt some of the notation in Gilon et al. (2019) for a more direct comparison.

Proof. Observe the following. First, no two P_i have type AB. If this were not the case, for example $P_1 = P_2 = AB$, then every donor of P_1 and P_2 could donate to the other patient. This contradicts with the set of feasible pairwise matches, which does not have this property for any pair of patient-donor groups. This implies that $d_i^1 \neq AB$ for all i. To see this, note that for each d_i^1, they can donate to two other patients. If for contradiction we had that for some i, $d_i^1 = AB$, then we would have that there are two AB patients because AB donors can only donate to AB patients. Furthermore, we have that for $i \in \{2, 3, 4\}$, $P_i \neq AB$. To see this, consider $P_2 = AB$ for contradiction. We have that $d_2^1 - d_4^2$ is a feasible pairwise exchange. As such, d_2^1 can donate to P_4, and since all donors can donate to $P_2 = AB$, then so can d_4^1. However $d_2^1 - d_4^1$ is not a feasible pairwise exchange, which is a contradiction. We can apply the same argument to the case where $P_3 = AB$ or $P_4 = AB$ (but note that it does not apply to P_1). Finally, observe that $P_i \neq O$ for $i \in \{2, 3, 4\}$. If this were not true, for example $P_2 = O$, then we would have that $d_1^1 = d_4^2 = O$. Given that O donors can donate to any agent, we would have that $d_1^1 - d_4^2$ would be a feasible exchange. This is a contradiction with the set of feasible exchanges given. The same argument applies to $P_3 = O$ or $P_4 = O$.

Using these facts, we proceed by considering various possible cases. First consider the case that $P_1 = AB$, thus any donor can donate to P_1. This implies $d_1 \neq O$, which follows because if $d_1 = O$, then we would have that any exchange is feasible for P_1 as P_1 can receive from any donor as they are AB and d_1^1 can donate to any patient as they are O. Recall that $d_1^1 \neq AB$, hence $d_1^1 \in \{A, B\}$. Consider the case that $d_1^1 = A$, and thus $P_2 = P_3 = P_4 = A$ (because none of these patients can be AB). Hence we must have that $d_2^2 \in \{A, O\}$ because $d_2^2 - d_3^1$ is feasible and $P_3 = A$, which means $d_1^1 - d_2^2$ is feasible and thus a contradiction. A similar proof applies when we instead assume that $d_1 = B$. As we have considered all the possible cases for choice of blood type of d_1 given $P_1 = AB$, and they all lead to contradictions, we cannot have that $P_1 = AB$.

Now consider the case where $P_1 = O$. As O patients can only receive from O donors, then we have that $d_2^1 = d_3^1 = d_4^1 = O$. This would imply that $d_3^1 - d_4^1$ is feasible, which is a contradiction. Hence we must have that $P_1 \in \{A, B\}$. Recall that we also have $P_2 \in \{A, B\}$.

Consider $P_1 = P_2 = A$. Thus we must have that $d_1^1, d_4^2 \in \{O, A\}$. If $d_1^1 = O$, then this would imply that $d_1^1 - d_4^2$ is feasible, which is a contradiction. If $d_1^1 = A$, then $P_4 = A$ (as it cannot be AB). Furthermore, if d_4^2 can donate to $P_2 = A$, then it can also donate to $P_1 = A$. And since d_1^1 can donate to P_4, we have that $d_1^1 - d_4^2$ is feasible, which is a contradiction. The same idea applies to $P_1 = P_2 = B$.

Now consider $P_1 = A$ and $P_2 = B$ (the same idea applies for $P_1 = B$ and $P_2 = A$). Note that $d_1^1, d_4^2 \in \{B, O\}$, because both donors can donate to $P_2 = B$. If $d_1^1 = d_4^2 = O$, then we would have $d_1^1 - d_4^2$ is feasible, which is a contradiction. If $d_1^1 = B$ and $d_4^2 = O$, then we have that $d_1^1 - d_4^2$ is feasible, which is also a contradiction. Consider $d_1^1 = O$ and $d_4^2 = B$. Because $d_1^1 - d_4^1$ is feasible and $P_1 = A$, it must be that $d_4^1 \in \{O, A\}$. If $d_4^1 = O$, then $d_4^1 - d_2^1$ would be feasible, which a contradiction. Hence it must be that $d_4^1 = A$. Given that $d_1^1 = O$ can donate to P_2, but d_2^2 cannot donate to $P_1 = A$ because $d_1^1 - d_2^2$ is not feasible, it must be that $d_2^2 \in \{B, AB\}$. First consider $d_2^2 = B$. In this case, note that $P_4 \in \{O, A\}$ because d_4^2 is compatible with P_2 but $d_2^2 = B$ is not compatible with P_4 as $d_2^2 - d_4^2$ is not feasible. If $P_4 = O$, then we must have that $d_2^1 = O$ as $d_2^1 - d_4^2$ is feasible. But this is a contradiction because we would have that $d_3^1 - d_2^1$ is feasible. If $P_4 = A$, then this would imply that $d_2^1 = \{A, O\}$. It must be that $d_2^1 = A$, as if $d_2^1 = O$ then we would have that $d_2^1 - d_3^1$ is feasible, which is a contradiction. Given this, we must have that $P_3 \in \{B, O\}$ so that $d_2^1 - d_3^1$ is not feasible (as $d_2^1 = A$). By our previous observation, it cannot be that $P_3 = O$. Hence if $P_3 = B$, we would get a contradiction as $d_3^2 - d_4^2$ would be feasible because d_3^2 is compatible with P_4 given the set of feasible exchanges, and $d_4^2 = B$ and $P_3 = B$ by assumption. Thus we cannot have that $d_2^2 = B$, so it must be that $d_2^2 = AB$. This would imply that $P_3 = AB$, which is a contradiction with out observation that $P_2, P_3, P_4 \neq AB$.

Now we consider the final case, that is $d_1^1 = d_4^2 = B$. This implies that $P_4 \in \{B, O\}$ as $d_1^1 - d_4^1$ is feasible and $d_1^1 = B$ by assumption. If $P_4 = B$, then $d_3^2 \in \{B, O\}$ as $d_2^3 - d_4^1$ is feasible and these are the only feasible blood types that can donate to $P_4 = B$. In either case, we would have that $d_2^2 - d_2^3$ is feasible because $P_2 = B$, which is a contradiction. Thus it must be that $P_4 = O$. However this contradicts our earlier observation that $P_4 \neq O$. As such it cannot be that $P_1 = A$ and $P_2 = B$ (or by an analogous argument that $P_1 = B$

and $P_2 = A$). As we have gone through all cases, and shown that there is no choice of blood types that are consistent with this set of feasible exchanges, then this set is not rationalizable. $\square$

A.3 Proof of Theorem 1

Individual rationality is clear. We focus on showing efficiency, stability, and strategy-proofness. Consider a matching M as the output of the mechanism for some preference profile in a given environment.

A.3.1 Stability

First observe that it is without loss of generality to consider coalitions of size two or three. Also note that the agents that may not have gotten their top choice include $O - *$ and $A - *$. Consider an $i \in \mathcal{I}_{A-*}$ that would strictly prefer to be matched by AB. Their possible exchanges include

1. $A - AB \leftrightarrow AB - A/O$: the other agent already gets their best match and thus cannot be strictly improve.

2. $A - AB \to AB - B \to j$ for some valid j: the second agent gets their best match.

Given the above reasoning, we can conclude that any deviating coalition cannot have a $j \in \mathcal{I}_{AB-*}$, and thus agents that can improve in a coalition, if any, cannot improve using an AB donor.

Note that there cannot be $i \in \mathcal{I}_{A-*}$ that would strictly prefer to be matched by A or O as if they did prefer this over the donor used in M, then they would have been matched with this donor in M.

Now consider an $i \in \mathcal{I}_{A-*}$ that would strictly improve by being matched by B. The only possible exchanges not already precluded is $A - B \leftrightarrow B - A$ or $A - B \leftrightarrow B - Y \to j$ for some Y and valid j: the second agent is already matched by their best donor in M, which is the case for all $k \in \mathcal{I}_{B-*}$. Given this reasoning, we can assume that no improving agent in a deviating coalition can belong to $\mathcal{I}_{B-*}$, and furthermore no agent can improve using their B donor as it requires the subsequent agent in the cycle to belong to $\mathcal{I}_{B-*}$ or $\mathcal{I}_{AB-*}$ and both we have shown can be precluded.

Now consider an $i \in \mathcal{I}_{O-*}$ that can be improved to O. This is not possible as they would have been matched in M via O if this were the case. Hence consider such an i that could be improved to A. The possible exchanges include the following:

1. $O-A \leftrightarrow A-O$: for the latter agent, if they strictly prefer O they would have been matched via it in M. Hence they do not strictly improve.

2. $O-A \rightarrow A-B \rightarrow j$ for some valid j: j must be of the form $B-Y$ or $AB-Y$, which we already showed cannot be strictly improved.

As for every agent that do not achieve their best exchange, the exchanges required to make them strictly improve have other agents that do not strictly improve (as they have already been matched with their top donor or will be matched using the same donor as in M), then there is no deviating coalition. Hence the matching is weak-core stable.

A.3.2 Strategy-proof

Before beginning, we address whether the dependence of the algorithm on whether $n_{A-B*} \geq n_{B-A*}$ can be strategically manipulated. Note that the top n_{B-A*} agents from $\mathcal{I}_{A-B*}$ given order Π will be matched via their top donor. Hence they have no incentive to lie. As every agent in $\mathcal{I}_{B-*}$ are matched with their top donor, they have no incentive to lie either. Thus any misreport from other agents will not affect whether $n_{A-B*} \geq n_{B-A*}$ holds.

Now we continue on studying reporting incentives assuming $n_{A-B*} \geq n_{B-A*}$. First we can observe that $A-O*$, $B-O*$, $b-b*$ for $b \in \{O, A, B\}$, and $B-A*$ all get their top choice and have no incentive to deviate. Furthermore, all $A-*$ agents are guaranteed their top two IR choices. Hence the only agents that do not get their top donor are $A-B, A*$ matched via A; $A-B, O*$ matched via O; and $A-B$ unmatched. If any other donor is put ahead of B, then A will be guaranteed to be matched by them. Hence there is no incentive to deviate as they will not be able to improve to their top donor instead of their second donor. Finally, $O-*$ can be seen to have no incentive to deviate by the Serial Dictator procedure, which uses a fixed priority order. Furthermore, all $O-*$ agents take the best option available to them when it is their turn.

A.3.3 Efficiency

We will note some useful observations.

Lemma 1. *1. S at step 7 is matchable.*

2. All $AB-O*$ are used by step 9, matched either via $B-AB \rightarrow AB-O \rightarrow A-O$, $B-AB \rightarrow AB-O \rightarrow B-O$, or $AB-O \leftrightarrow O-AB$.

3. All $AB - A*$ are used by step 9, matched via $B - AB \to AB - A \to A - B$.

4. $B - AB*$ is matched by either the previous two, or $AB - B \leftrightarrow B - AB$.

Proof. For the first claim, note that $n_{AB-*} > n_{B-AB*}$ and $B-AB$ can match via $B-AB \leftrightarrow AB - O*$, $B - AB* \leftrightarrow AB - B*$, or $B - AB* \to AB - A* \to A - B*$. Note that the last match is always possible as $AB - A < A - B*$ (since the latter includes $A - AB*$ that dropped their AB). Furthermore $A - B*$ are all matched as we chose exactly n_{AB-A*} of them. Finally, any unmatched $AB - *$ can be matched with each other as there are at least two $AB - AB*$.

For the second claim, if there were some $AB - O*$ not matched, then some $O - AB*$ would have matched with them. This is because $n_{AB-O*} < n_{O-AB*}$, and there must be at least one left. To see why there can only be one left, note that these are the only ways for $O - AB$ to match: $O - AB \leftrightarrow AB - O$, $O - AB \to AB - A \to A - O$, and $O - AB \to AB - A \to B - O$ (the cases with $AB - O$ can be thought of as the first case). However, we must ensure that all $B - AB*$ are matched, and we've already committed the matches via $B - AB \leftrightarrow AB - B$ and $B - AB \to AB - A \to A - B$. These are all other agents that $O - AB$ would require to match with, apart from $AB - O$ directly, hence this is the only option left. This shows our claim. Note that we've also shown the last two claims. □

Given this, observe that every $AB - *$ is used, and furthermore there is always some agents that requires one of them. Thus if there exists a Pareto improving matching, such agents cannot be matched with agents that do not include some (not necessarily the same) $AB - *$. This is the case for $B - AB*$ and any $AB - A*$ they're matched with, and so now we will show that's also the case for those matched with $AB - O*$. If they are matched $O - AB*$, then this is necessary by the same reasoning in the proof of the lemma.

We now use the same logic as serial dictator. Note that all $O - *$, to strictly improve, require the use of $A - O$, $B - O$, or $AB - O$. Let i be the highest priority $O - *$ that can be improved. Hence everyone above them cannot be improved. If i were not matched via their more preferred option at this stage, then it must be because doing so would make others worse off. Hence it is not possible for there to be an $O - *$ that can be strictly improved. The only other agent that could potentially strictly improve is any $A - *$ that would have preferred to use AB. However, as we noted earlier, any option they have $AB - A/B/O$ would be at the expense of another agent. Thus we conclude that no agent can be strictly improved, and thus the matching is efficient.

A.4 Proof of Proposition 3

Proof. First consider the case where there is an AB patient. If the AB patient has a donor compatible with some other patient, then they must be pointing to someone. Since anyone can donate to AB patients, then everyone is pointing to AB. Hence there is a two-cycle. If the AB patient has no donor that can feasibly donate to another patient, then they are not part of any cycle.

Now consider the case where there is no AB patient. This means that any cycle cannot utilize AB donors, as AB donors can only donate to AB. Hence assume that there are no AB donors. Assume there is no two cycle but there is a cycle of length n.

First we will show that there cannot be an O donor as an patient's top choice, nor an O patient, for any agent in a cycle. Assume for contradiction that there is an agent i_1 that points with an O donor in the cycle. Hence they point to every agent. Since $i_n \to i_1$, hence $i_1 \leftrightarrow i_n$. Since there are no O donors, and only O donors can donate to O patients, then there can be no O patients.

We proceed by assuming that there are only A and B donors and patients, and consider different cases based on the number of agents n in the cycle.

If there are only two agents, and thus all these agents are in the cycle, then this is a contradiction.

If there are at least five agents in the cycle, denoted $i_1 \to i_2 \to i_3 \to ...$, then observe we can't have the any three consecutive agents have the same type. For example, if $i_1, i_2, i_3 \in \mathcal{I}_A$, then $i_2 \to i_1$ and thus there is a two cycle, which is a contradiction.

First consider the case where $i_1, i_2 \in \mathcal{I}_A$. This implies that $i_3 \in \mathcal{I}_B$. If the cycle is of length three, then we are done. If $i_3 \to i_4 \in \mathcal{I}_A$, then $i_3 \to i_2$, which introduces a two cycle. Hence $i_4 \in \mathcal{I}_B$. If the cycle is of length four, then we are done. If $i_4 \to i_5 \in \mathcal{I}_B$, then $i_4 \to i_3 \in \mathcal{I}_B$, another contradiction. If $i_4 \to i_5 \in \mathcal{I}_A$, then $i_4 \to i_2 \in \mathcal{I}_A$ and $i_2 \to i_4 \in \mathcal{I}_B$ since $i_2 \to i_3 \in \mathcal{I}_B$. This final contradiction shows that i_1, i_2 cannot both be in $\mathcal{I}_A$. A similar argument holds for $i_1, i_2 \in \mathcal{I}_B$.

Now consider the case where $i_1 \in \mathcal{I}_A$ and $i_2 \in \mathcal{I}_B$. If there are only three or four agents, then we are done (assuming in the latter case there is no three consecutive agents of the same type). Assume there are at least five agents: $i_1 \to i_2 \to i_3 \to i_4 \to i_5$. If $i_3 \in \mathcal{I}_A$ and $i_4 \in \mathcal{I}_B$, then $i_3 \to i_2$, giving a contradiction. By the previous argument on consecutive types, it cannot be that $i_3, i_4 \in \mathcal{I}_B$. If $i_3, i_4 \in \mathcal{I}_A$, then $i_5 \in \mathcal{I}_B$, otherwise there will be three agents of consecutive types. Then $i_4 \to i_2$ and $i_2 \to i_4$ gives a two cycle and thus a contradiction. Now consider $i_3 \in \mathcal{I}_B$ and $i_4 \in \mathcal{I}_A$, then $i_1 \to i_3$ and $i_3 \to i_1$, another contradiction. A similar argument applies for $i_1 \in \mathcal{I}_B$ and $i_2 \in \mathcal{I}_A$.

If there are exactly three agents, it is clear that the cycles must be of the following form: $A \to B \to A$, or $B \to A \to B$. Note that, for example, the latter is equivalent to $A \to B \to B$. However then it must be that there is a two-cycle given by $A \leftrightarrow B$ in both cases, thus this is not possible. If there are four agents, since there can be no three consective agents, it must be that cycles are either of the form: $A \to B \to A \to B$, or $A \to B \to B \to A$. However neither are possible as there is a two cycle, which is a contradiction.

$\square$

B Miscellaneous

B.1 Assumptions on patient distribution

As in Roth et al. (2007a), given the three PRA levels we can calculate the probability of tissue-type incompatibility for a random pair is approximately 20%. Thus the inequality, using the same blood type distribution as in Roth et al. (2007a), can be determined by considering the probability of a certain patient-donor pair $X - Y$ having being incompatible (that is they are not blood type and tissue type compatible): $(0.14 + 0.33) \cdot p \leq 0.14 \leq (0.14 + 0.33 + 0.48) \cdot p$, where normalize the total number of agents to be unit and let p be some probability of tissue-type incompatibility. Hence we have that $p \in [0.15, 1]$ approximately, which holds for $p = 0.2$. Note that for simplicity we assume independence between the blood types of X, Y, as well as the probability of tissue-type incompatibility.